

the German version, moreover, the term "Tierarten" in Art. 53(b) EPC 1973 was replaced by "Tierrassen" to bring it into line with the Biotech Directive and R. 23c(b) EPC 1973 (MR/2/00, 45).

In addition, the exclusion of medical methods which had previously been enshrined in Art. 52(4) EPC 1973 was transferred to Art. 53(c) EPC. These methods had formerly been excluded from patentability through the legal fiction that they were not susceptible of industrial application. However, since medical methods are excluded from patentability mainly in the light of public health considerations, it appeared appropriate to include these inventions also under exceptions to patentability (MR/2/00, 45).

1.2. Basic principles

No general exclusion of inventions in the sphere of animate nature can be inferred from the EPC (T 49/83, OJ 1995, 545). It was held in T 356/93 (OJ 1995, 545) that seeds and plants per se should not constitute an exception to patentability under Art. 53(a) EPC 1973 merely because they represented 'living' matter, or on the ground that plant genetic resources should remain the 'common heritage of mankind'.

The case law indicates that any exceptions to patentability must be narrowly construed. In respect of Art. 53(a) EPC, see T 356/93 (OJ 1995, 545) and T 866/01, but also T 1374/04 (OJ 2007, 313); in respect of Art. 53(b) EPC, see T 320/87 (OJ 1990, 71), T 19/90 (OJ 1990, 476) and T 315/03 (OJ 2006, 15); regarding Art. 53(c) EPC (Art. 52(4) EPC 1973) see T 144/83 (OJ 1986, 301), T 385/86 (OJ 1988, 308) and G 1/04 (OJ 2006, 334).

Given the ratio legis of the individual provisions, however, this narrow interpretation produces different results: a claim which embraces plant/animal varieties, but does not claim them individually, is not excluded from patentability under Art. 53(b) EPC (G 1/98, OJ 2000, 111; T 19/90, OJ 1990, 476; T 315/03, OJ 2006, 15). According to the established case law of the boards of appeal, a method claim falls under the prohibition of Art. 53(c) EPC if it includes at least one feature defining a physical activity or action that constitutes a method step for treatment of the human or animal body by surgery or therapy (see in this chapter I.B.4.2.). By contrast, several method steps are required to define a diagnostic method within the meaning of Art. 53(c) EPC owing to the inherent and inescapable multi-step nature of such a method (G 1/04, OJ 2006, 334). In T 19/90 it was also stated that the object and purpose of the law (ratio legis) was not merely a matter of the legislator's intention at the time when the law was adopted, but also of their presumed intention in the light of changes in circumstances which had taken place since then.

2. Breaches of "ordre public" or morality

In the past, this issue has arisen mainly in connection with biotechnological inventions.

Art. 53(a) EPC is supplemented by R. 28 EPC, which sets out four categories of biotechnological inventions excluded from patentability under that article. R. 28 EPC is part of the chapter on "Biotechnological inventions" inserted into Part II of the

Implementing Regulations, and is identical to Art. 6(2) of the Biotech Directive. The Biotech Directive was thereby transposed into European patent law.

Where an invention falls under one of the categories set out in R. 28 EPC, it is by its very nature non-patentable and there is no need additionally to consider Art. 53(a) EPC. However, if it does not fall under one of those categories, it must be examined more closely under Art. 53(a) EPC (T 315/03, OJ 2006, 15; see also G 2/06, OJ 2009, 306).

R. 29 EPC governs the patentability of the human body and its components. Under R. 29(2) EPC (R. 23e (2) EPC 1973), an element isolated from the human body or otherwise produced by a technical process, including the sequence or a partial sequence of a gene, may constitute a patentable invention, so that an invention falling under this category is not excluded from patentability pursuant to Art. 53(a) EPC (T 272/95 of 23 October 2002; see also T 1213/05).

2.1. Rule 28 EPC

R. 28 EPC, which sets out four non-patentable categories, entered into force on 1 September 1999 as R. 23d EPC 1973. It applies equally to applications pending on, i.e. filed before, the date of its entry into force (G 2/06, T 315/03).

2.1.1 Scope of exception under Rule 28(c) EPC

Under R. 28(c) EPC (R. 23d(c) EPC 1973), European patents may not be granted in respect of biotechnological inventions which concern "uses of human embryos for industrial or commercial purposes". The expression "uses for industrial or commercial purposes" in R. 28(c) EPC covers the making of the claimed product, even where it was intended to use that product for further research (G 2/06).

In G 2/06, the Enlarged Board considered an invention concerning, inter alia, human embryonic stem cell cultures which at the filing date could only be prepared by a method which necessarily involved the destruction of the human embryos from which they were derived, that method not being part of the claims.

When interpreting R. 28(c) EPC, the Enlarged Board also looked at Art. 6(2) Biotech Directive because the aim of introducing R. 26 to 29 EPC was to align the EPC to that directive. It observed that neither the EU legislator nor the EPC legislator, unlike their German and UK counterparts, had chosen to define the term "embryo". It assumed that they had been aware of the definitions in national laws on regulating embryos and deliberately refrained from defining the term. Given the aim of protecting human dignity and preventing the commercialisation of embryos, it could only presume that "**embryo**" was not to be construed restrictively and therefore rejected the appellant's argument that "embryo" was to be given a very specific interpretation as meaning embryos of 14 days or older, in accordance with usage in the medical field.

The appellant had argued that, for the R. 28(c) EPC prohibition to apply, the use of human embryos had to be claimed. The Enlarged Board disagreed, observing that R. 28(c) EPC

did not mention claims, but rather referred to "invention" in the context of its exploitation. What had to be looked at was not just the explicit wording of the claims but the technical teaching of the application as a whole as to how the invention is to be performed. The Enlarged Board thus answered the second point referred as follows: R. 28(c) EPC forbids the patenting of claims directed to products which – as described in the application – at the filing date could be prepared exclusively by a method which necessarily involved the destruction of the human embryos from which the said products are derived, even if the said method is not part of the claims.

In T. 522/04, the contested claim was directed to a method of proliferating *in vitro* a clonal population of neural crest stem cells of mammalian origin. This clearly included cells of human origin. Since the only teaching of how to prepare human neural crest stem cell cultures was the use (involving their destruction) of human embryos, the board concluded that at the filing date human neural crest stem cells could be prepared exclusively by a method which necessarily involved the destruction of human embryos – with the inescapable result that the invention fell under the prohibition of Art. 53(a) EPC combined with R. 28(c) EPC.

In T. 2221/10 claims 1 and 2 of the appellant's sole request referred to methods of maintaining human embryonic stem cells (hES cells) in culture in an undifferentiated state, while claim 5 referred to a cell culture comprising hES cells. The appellant argued that methods using commercially or otherwise publicly available hES cell lines were not excluded from patentability because no de novo destruction of human embryos was necessary to perform them.

The board disagreed. It referred to G. 2/06 and noted that, according to that decision, all steps preceding the claimed use of hES cells which are a necessary precondition for carrying out the claimed invention have to be considered. In this respect the Enlarged Board of Appeal had neither made a distinction between steps which have been carried out by the inventor or by any other person, nor between steps which took place in direct preparation of the experiments leading to an invention and steps having taken place at a point in time further remote from those experiments. The board thus decided that inventions which make use of hES cells obtained by de novo destruction of human embryos or of publicly available hES cell lines which were initially derived by a process resulting in the destruction of the human embryos are excluded from patentability under the provisions of Art. 53(a) EPC in combination with R. 28(c) EPC. It noted that its decision was in line with ECJ judgment C-34/10.

In T. 1441/13 claim 1 of the main request was directed to a method for obtaining polypeptide-secreting cells. The method required the use of a culture of primate pluripotent stem cells (pPS) which, according to the description in the application, included human embryonic stem (hES) cells. The board considered that at the relevant date of the patent in suit, the known and practised method for achieving cultures of hES cells, i.e. the starting material of the method of claim 1, included preceding steps that involved the destruction of human embryos. Thus, the main request was not allowable under Art. 53(a) EPC and R. 28(c) EPC.

2.1.2 Scope of exception under Rule 28(d) EPC

R. 28(d) EPC (R. 23d(d) EPC 1973) provides that European patents should not be granted in respect of processes for modifying the genetic identity of animals which were likely to cause them suffering without any substantial medical benefit to man or animal, and also animals resulting from such processes.

One of the most important cases dealing with interpretation not only of R. 23d EPC 1973, but also of Art. 53(a) EPC, concerned a patent relating to **transgenic animals having an increased probability of developing cancer**. The application resulted in two groundbreaking board decisions: T. 19/90 (OJ 1990, 476, see below), in which the matter was remitted to the department of first instance, and T. 315/03 (OJ 2006, 15), in which a patent was granted on the basis of new claims.

Having addressed the relationship between R. 23d EPC 1973 (R. 28 EPC) and Art. 53(a) EPC (see also in this chapter I.B.2.), the board in T. 315/03 turned to the test under R. 23d(d) EPC 1973 (R. 28(d) EPC). This test required **only** three matters to be considered: animal suffering, medical benefit and the necessary correspondence between the two in terms of the animals in question. The board observed that the test was to be applied to ensure that a patent extended only to those animals whose suffering was balanced by a medical benefit. The level of proof was the same for both animal suffering and substantial medical benefit, namely likelihood (see also T. 1262/04 of 13 July 2012).

In T. 1553/15 the board held that contrary to the transgenic mouse of T. 19/90, which opened up new research avenues in the field of oncology at the cost of the suffering of a limited number of animals, the board considered that the benefit to mankind brought by the present invention was not such as to weigh up against the suffering of animals which was necessary to produce the claimed pharmaceutical composition. The new pharmaceutical composition did not open up new avenues in the treatment of the claimed diseases, and animal suffering was not limited to a given number of animals needed for testing but rather was always present and involved a considerable number of animals every time the composition was produced. There were also alternatives to the claimed pharmaceutical composition available.

2.1.3 Test date

In T. 315/03, the board observed that the relevant date for applying the test under R. 23d EPC 1973 (R. 28 EPC) was, as for all patentability criteria, the date of filing or priority. Nevertheless, evidence becoming available later could be taken into account, provided it related to the situation on that date.

In G. 2/06, the Enlarged Board arrived at the same basic result. When assessing whether a claim contravenes R. 28(c) EPC, technical developments which became publicly available only after the filing date cannot be taken into consideration. Any other conclusion would lead to legal uncertainty, and risk being to the detriment of any third party who later provided an innocuous way to carry out the invention. The Enlarged Board thus concluded in the case before it that it is irrelevant that, after the filing date, the same products could

be obtained without having recourse to a method necessarily involving the destruction of human embryos.

2.2. Article 53(a) EPC

If an invention is not excluded from patentability because it falls under one of the four categories in R. 28 EPC (see above), it must be then examined whether it is covered by the exception to patentability under Art. 53(a) EPC.

2.2.1 Test date

Assessment of an Art. 53(a) EPC objection was to be made as of the filing or priority date of the patent or application in suit. Evidence arising thereafter might be taken into account provided it was directed to the position at that date (T 315/03).

2.2.2 Assessment of an objection according to Article 53(a) EPC

In T 19/90 (OJ 1990, 476), the board expressed the view that in cases, like the one before it, of genetic manipulation of animals by the insertion of an activated oncogene, there were compelling reasons to consider Art. 53(a) EPC 1973. As this had not been done at the level of the first instance, the board remitted the case to the examining division with the instruction to carry out a careful **weighing up** of the suffering of animals and possible risks to the environment on the one hand, and the invention's usefulness to mankind on the other, before deciding whether to grant or refuse the patent application (see also T 1262/04 of 13 July 2012).

a) Difference from test under Rule 28(d) EPC

In T 315/03 (OJ 2006, 15) the board held that, in animal manipulation cases, the test in T 19/90 (OJ 1990, 476) was appropriate. This differed in several respects from the test in R. 23d(d) EPC 1973 (R. 28(d) EPC), most importantly by allowing matters other than animal suffering and medical benefit to be taken into account. Whereas the R. 23d(d) EPC 1973 test only required a likelihood of animal suffering, however minor, and a likelihood of achieving a medical benefit, the test in T 19/90 required a "careful weighing up" of the matters to be balanced. This clearly allowed an appraisal of animal suffering, of environmental risks and of the feasibility of using non-animal alternatives.

A wider range of benefits to mankind than the substantial medical benefit found in R. 23d(d) EPC 1973 could also be considered. Since the test in T 19/90 was "mainly" the basis of assessment, further arguments as to the appropriate standard of morality or "ordre public" could additionally be considered, but all arguments had to be supported by evidence.

In T 315/03 the board found that claims directed to rodents failed the R. 23d(d) EPC 1973 test and had therefore to be refused, as the patent disclosed a likelihood of animal suffering but not a likelihood of medical benefit when applying the invention to this generic class of